

EX. NO: 11 — IMPLEMENTATION OF DIGITAL SIGNATURE STANDARD

Aim:

To implement the Digital Signature Standard (DSS) using C and verify a digital signature.

Algorithm:

1. Read the message and private key.
2. Generate a digital signature using the private key.
3. Generate the corresponding public key.
4. Verify the signature using the public key.
5. Display whether the digital signature is valid or invalid.

Program:

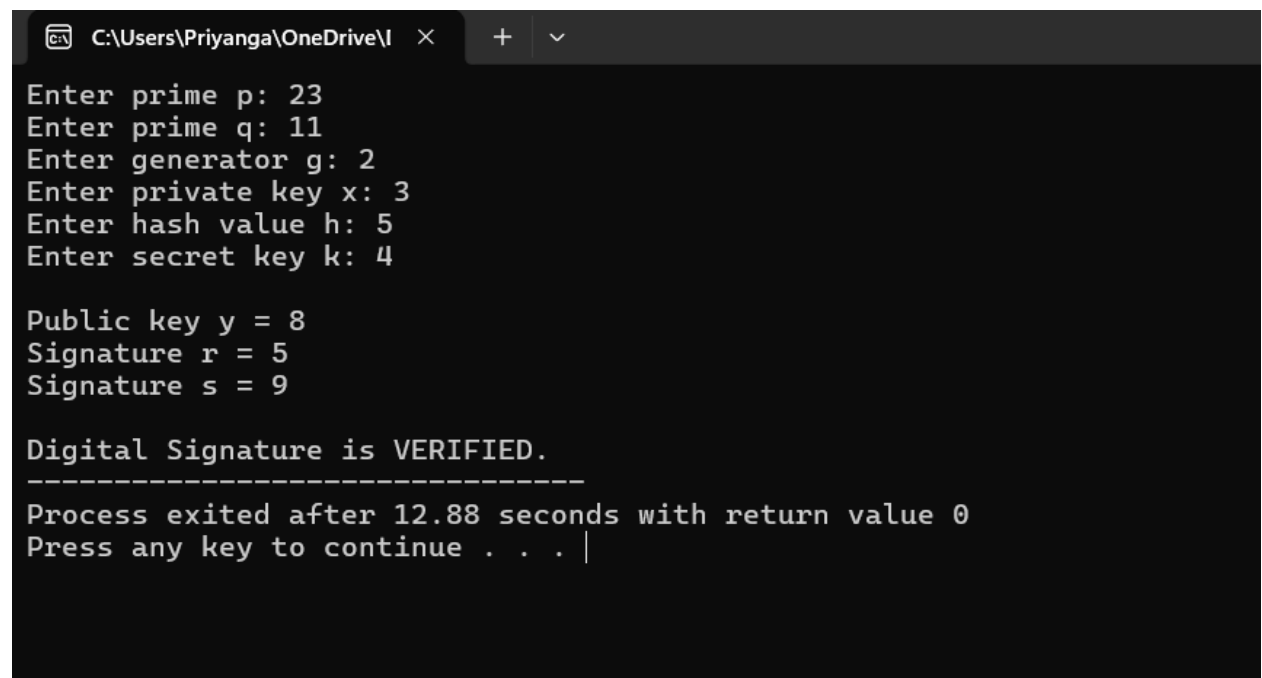
```
#include <stdio.h>

long long powerMod(long long a, long long b, long long m)
{
    long long result = 1;
    while(b > 0)
    {
        result = (result * a) % m;
        b--;
    }
    return result;
}

int main()
{
    long long p, q, g, x, h, k;
```

```
long long y, r, s;
printf("Enter prime p: ");
scanf("%lld", &p);
printf("Enter prime q: ");
scanf("%lld", &q);
printf("Enter generator g: ");
scanf("%lld", &g);
printf("Enter private key x: ");
scanf("%lld", &x);
printf("Enter hash value h: ");
scanf("%lld", &h);
printf("Enter secret key k: ");
scanf("%lld", &k);
y = powerMod(g, x, p);
r = powerMod(g, k, p) % q;
s = (h + x * r) % q;
printf("\nPublic key y = %lld", y);
printf("\nSignature r = %lld", r);
printf("\nSignature s = %lld", s);
if ((s - h + q) % q == (x * r) % q)
    printf("\n\nDigital Signature is VERIFIED.");
else
    printf("\n\nDigital Signature is INVALID.");
return 0;
}
```

Output:



```
C:\Users\Priyanga\OneDrive\I  X  +  v
Enter prime p: 23
Enter prime q: 11
Enter generator g: 2
Enter private key x: 3
Enter hash value h: 5
Enter secret key k: 4

Public key y = 8
Signature r = 5
Signature s = 9

Digital Signature is VERIFIED.
-----
Process exited after 12.88 seconds with return value 0
Press any key to continue . . . |
```

Result:

Thus, the Digital Signature Standard technique was implemented and the digital signature was verified successfully.